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CAEP POSITION STATEMENT & GUIDELINES

Canadian Emergency Department Triage and Acuity Scale (CTAS) Guidelines 2025

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Introduction

The Canadian Triage and Acuity Scale (CTAS) was established in 1991 to standardize triage in emergency departments (ED) across Canada and has since evolved to meet the dynamic needs of the healthcare system [1]. CTAS assigns patients to one of five levels based on presenting symptoms and vital signs, helping to prioritize care according to urgency. It is now widely used in EDs, urgent care centers, nursing stations, and pre-hospital paramedicine [2]. Recognizing the need for consistent and evidence-based triage practices nationwide, the Canadian Association of Emergency Physicians (CAEP) and the National Emergency Nurses Association (NENA) partnered to create the CTAS National Advisory Committee. Members are selected based on their clinical expertise, leadership roles, and contributions to emergency care and triage education. The committee's governance structure ensures balanced representation from key stakeholder groups, facilitates collaborative decision-making, and maintains accountability through regular reporting to the parent organizations. The committee is responsible for maintaining and updating the CTAS guidelines, developing standardized educational materials, and serving as the national authority on triage practice to ensure consistency and integrity in how CTAS is applied across healthcare settings.

Although CTAS has historically followed a structured 4-year revision cycle with updated teaching packages and participant manuals, the last major guidelines update occurred in 2016 (published in 2017) [2] without accompanying revisions to its educational resources. An update to these guidelines is therefore long overdue particularly amid the ongoing national shortage of experienced emergency nurses and reliance on CTAS materials dating back to 2013. This situation underscores an urgent need for new educational tools that accurately reflect today's clinical practice and support the safe, effective use of the CTAS framework.

Embracing a continuous quality improvement approach, the National Advisory Committee integrates regular feedback from emergency nurses, physicians, administrators, and instructors to keep the guidelines current. This paper outlines the most recent updates to CTAS, highlighting key changes to the guidelines while incorporating the principles of fairness, representation, and accessibility in the updated

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bilingual content. It introduces a modernized educational framework designed to support consistent and effective triage across diverse clinical settings.

Methods

The National Advisory Committee employs a systematic, iterative process that integrates evidence-based review of clinical literature and feedback from triage educators and frontline users on clarity, applicability, and usability of existing guidelines to ensure revisions align with contemporary standards of care and real-world practice. As part of this process, the committee conducted a comprehensive analysis of the 169 CEDIS complaints and their associated modifiers. In the 2016 CTAS guidelines, significant changes to the CEDIS complaint list were implemented with the introduction of several new modifiers, bringing the total to over 400. Each modifier was scrutinized for opportunities to simplify its application based on feedback from frontline nurses and instructors. Subsequently, the National Advisory Committee cross-referenced expert opinions from clinicians, researchers, and specialists with current literature on triage assessment, prioritization, and operational processes to ensure that the guidance provided to triage nurses is both up-to-date and clinically relevant. Moreover, this process specifically addresses the nuanced needs of diverse patient populations, including pediatric, geriatric, rural and remote communities, and individuals with unique healthcare challenges, such as sickle cell disease or immunocompromised states.

Decisions within the CTAS National Advisory Committee were made through a collaborative and consensus-driven process. Committee members engaged in regular meetings where evidence reviews, member feedback, and clinical considerations were discussed in detail. When differing opinions or discrepancies arose, the committee employed open dialog and deliberation to explore various perspectives. Subject matter experts were consulted to opine on contentious issues and provide recommendations. If consensus could not be reached through discussion, the committee used formal voting procedures to resolve disagreements.

Revisions

Revisions to the CTAS guidelines incorporate the latest clinical recommendations, evidence-based practices, and expert consensus to ensure a more precise and consistent triage process. The objective is to improve patient outcomes by refining the assessment of clinical severity, in conditions, such as chest pain, respiratory distress, trauma, and pediatric emergencies, through updated criteria for triage level assignment. These refinements ensure that the most critically ill patients are prioritized appropriately.

This update focuses on four key areas:

1. Canadian Emergency Department Information Systems (CEDIS) Complaints: The revised CTAS guidelines maintain consistency with the 2016 update for all 169 CEDIS complaints. However, select complaints have been reclassified to improve the accuracy of chief complaint placement. For instance, sexual assault, head injury, and traumatic back/spine injury are now categorized under Trauma, while concerns, such as patient welfare, general weakness, and social problems, are reassigned to the General category (renamed from the previous General Minor category).
2. Modifiers Update: Feedback from frontline clinicians and instructors revealed that the terminology of "first-order," "second-order," and "special' modifiers was ambiguous and insufficiently specific. In response, the committee renamed first-order modifiers as "Primary Modifiers" to better reflect their association with key assessment parameters (airway, breathing, circulation, disability, pain, and mechanism of injury). Additionally, the frailty modifier, introduced in the 2016 guidelines, has been expanded to encompass all CEDIS complaints and is now classified as a Primary Modifier [3]. Primary Modifiers have been added for complaints including sensory loss/paresthesias, diplopia, visual disturbances, pruritus, rash, and wound check. The previously distinct second-order and special modifiers, which are related to the specific presenting complaint and the patient's medical history, have been unified under the term "Complaint Specific Modifiers." Furthermore, a new CTAS 2 level has been introduced for patients with frailty who are at higher risk if care is delayed; currently, this applies exclusively to neonates aged seven days or less and premature infants with a corrected age of less than 3 months.

A summary of significant changes to the modifiers includes:

- 85 Complaint Specific Modifiers have been retired.
- 4 CEDIS Complaints have had Primary Modifiers removed.
- 5 CEDIS Complaints have been assigned a new lowest possible CTAS score.
- 244 new Complaint Specific Modifiers have been designated.

- 6 CEDIS Complaints have had Primary Modifiers added.
- A new CTAS level option has been incorporated into the frailty modifier, with frailty now applied to all 169 CEDIS complaints.

These changes simplify modifier terminology and improve clarity, enabling more precise and consistent triage decision-making. The full CTAS guidelines including a complete list of changes can be seen in the supplement.

3. Enhanced Documentation Requirements: Documentation standards have been updated to ensure that triage decisions are supported by clear, standardized information. This enhancement improves quality control of the triage process and ensures that decisions can be transparently reviewed and justified.
4. Training and Education Initiatives: Accompanying the guidelines revision is an updated training program for all triage personnel. This initiative includes workshops for current instructors (with an online assessment to confirm competency), a new CTAS triage education course available to all triage nurses, and comprehensive reference materials including a revised guidelines document, updated education slides, participant manuals, and the Complaint Oriented Triage (COT) tool. Future plans include the integration of e-learning models. Training for the updated guidelines commenced in February 2025.

Importantly, these updates will have implications for facilities using electronic triage decision support systems, as computer-based triage tools will require modifications to be consistent with the current guidelines. Collectively, these revisions represent a significant advancement in standardizing triage procedures and optimizing patient care in alignment with modern clinical practice.

Limitations

While the recent CTAS updates represent a significant advancement, several limitations should be acknowledged. First, the iterative revision process, though comprehensive, relies on heterogenous clinical evidence depending on the presenting complaint which can range from expert opinion to association guidelines based on high-quality prospective research. In cases of emerging conditions or rare presentations, even expert opinion was not available in the literature. Second, despite efforts to incorporate diverse patient populations, some unique or complex cases may still challenge the applicability of standardized triage criteria. Third, the implementation of updated guidelines and educational materials depends on effective dissemination and uptake across varied healthcare settings, which may be inconsistent due to resource constraints or training gaps, particularly in rural and remote areas. Finally, ongoing evaluation is needed to assess the real-world impact of these changes on triage accuracy, patient outcomes, and workflow efficiency.

Next steps

With the new CTAS guidelines, NENA has initiated a comprehensive nationwide training program for CTAS instructor trainers, CTAS instructors, and triage nurses. Concurrently, updates and validation of electronic databases and clinical decision support tools are underway to align with the revised guidelines, and a French translation of both the guidelines and associated educational materials is in progress. Additionally, the National Advisory Committee is analyzing data from a recent national survey of ED leaders, triage nurses, and physicians to evaluate the current triage landscape, with results to be shared when available. Finally, the updated CTAS guidelines are serving as the foundation for revising the Prehospital Canadian Triage and Acuity Scale (Pre-CTAS) to better support pre-hospital settings and providers.

Conclusions

The updated CTAS guidelines are anticipated to streamline the triage process by enhancing both efficiency and accuracy. By aligning clinical practice with the latest evidence and expert consensus, these revisions are poised to strengthen decision-making in emergency settings, ultimately ensuring that patients receive the highest standard of care nationwide.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s43678-025-00996-1.

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Declarations

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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